

Integrated Planning for Transition to Low-Carbon Distribution System With Renewable Energy Generation and Demand Response

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Abstract—This study presents an integrated methodology that considers renewable distributed generation (RDG) and demand responses (DR) as options for planning distribution systems in a transition towards low-carbon sustainability. It is assumed that demand responsiveness is enabled by real-time pricing (RTP), and the problem has been formulated as a dynamic two-stage model. It co-optimizes the allocation of renewables [including wind and solar photovoltaic (PV)], non-renewable DG units (gas turbines) and smart metering (SM) simultaneously with network reinforcement for minimizing the total economic and carbon-emission costs over planning horizons. The behavior compliance to RTP is described through a nodal-based DR model, in which the fading effect attended during the load recovery is highlighted. Besides, uncertainties associated with renewable energy generation and price-responsiveness of customers are also taken into account and represented by multiple probabilistic scenarios. The proposed methodology is implemented by employing an efficient hybrid algorithm and applied to a typical distribution test system. The results demonstrate the effectiveness in improving the efficiency of RDG operations and mitigating CO₂ footprint of distribution systems, when compared with the conventional planning paradigms.

Index Terms—Distribution system planning, low-carbon characteristics, real-time pricing (RTP), renewable distributed generation (RDG), smart metering (SM), uncertainty.

NOMENCLATURE

A. Sets

Ω_F	Set of right-of-ways.
Ω_a	Set of conductor types.
$\Omega_G^{\text{rdg}}/\Omega_D$	Set of RDG connection/load buses.

$\Omega_G^{\text{gt}}/\Omega_G^{\text{w}}/\Omega_G^{\text{pv}}$ Set of candidate buses for gas/wind/PV allocation.

Ω Set of all the system buses.

B. Indices

$i, j/g/k$	Index of buses/DG buses/load points.
t	Index of time periods.
λ	Index of planning years.
s	Index of synthetic scenarios.
a	Index of feeder conductors.

C. Parameters

D/D_q	Total nodal active/reactive demand.
$D_0^{\text{re}}/D^{\text{ure}}$	Forecasted responsive/unresponsive demand.
ε	Elasticity.
ϑ	Ratio of payback to reduced consumption.
ρ_0	Initial fixed offering tariff.
ρ^{gsp}	Day-ahead market price of electricity.
ρ^{gt}	Production cost per kWh of energy by gas turbines.
ρ^{los}	Cost per kWh of network energy losses.
$\bar{\omega}$	Fading coefficient.
TR	Duration of load recovery process.
TY	Total planning horizons.
TL	Equipment lifespan.
NS	Total number of considered scenarios.
d	Discount rate.
C	Capital cost of the equipment.
l	Length of feeders in km.
n^{hd}	Number of households in load points.
C^{ct}	CO ₂ emission tax rate (\$/t).
eb	Embodied CO ₂ in per unit of system components.
eg^{gsp}	Emission factor of the main grid (kg CO ₂ /kWh).

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PR	Occurrence probability of scenario.
r_t	Duration of time period t (one hour).

D. Variables

$\tilde{D}^{re}$	Single-period modified demand.
κ	Penetration rate of SM.
$P^{cov}(t, t')$	Payback demand from period t to t' .
$P_g^w / P_g^{pv} / P_g^{gt}$	Available power output by DWG/PV/gas turbines.
$n_{g,\lambda}^w / n_{g,\lambda}^{pv} / n_{g,\lambda}^{gt}$	Number of wind turbines/PV units/gas turbines.
$\tau_{ij,a,\lambda}$	Decision variable of conductor selection.
$P_{i,s}^{cur}$	Curtailed RDG power.
$P_{i,s}^{gsp} / Q_{i,s}^{gsp}$	Active/reactive power supplied by main grid.
Z	Residual value of the equipment.
$V_{i,s}$	Nodal voltage.
$I_{ij,s}$	Current in feeder ij .
$\rho_{k,s}$	Offering real-time tariff.

I. INTRODUCTION

THE emission of CO₂ by human beings is acknowledged as one paramount factor of causing the current climate change.

The electric power sector contributed approximately 41% of the global CO₂ emissions in the year of 2012 [1] and even higher in thermal-dominated countries, such as China. To tackle this challenge, power industry decarbonization becomes crucial and necessary [2].

Distribution systems serve as an indispensable linkage in the modern power system, whereby the generated electricity is managed and delivered to end-users. Once for a long time, as all the energy has to be purchased from the designated entities in the market such as local transmission companies, there are neither economic incentives nor technical conditions for distribution companies (DISCOs) to consider CO₂ mitigation from distribution level. However, with the deregulation of the power sector and taxation on CO₂ emissions in many countries, renewable distributed generation (RDG) offers DISCOs flexible alternative options to meet load growth but with low environmental burdens, which prevails and brings new opportunities [3].

The performance of distribution systems can be largely altered with the integration of RDG. Therefore, to facilitate RDG for their benefits, a variety of planning strategies have been suggested in the literature [4]–[14]. In [4], a probabilistic heuristic method is proposed for determining the optimal mix of different RDG technologies to minimize system energy losses. For the same objective, an analytical technique to the planning is described in [5], and a multi-period optimal flow analysis is carried out by [6] in which the smart control schemes are considered. With multi-objective programming, the issue of maximizing wind power integration in networks is discussed in [7].

Similar studies have also been pursued in [8] and [9] but with the extension to cater for the economic criteria. In the studies above, the network capacity is normally assumed to be unchanged and reinforcement along with RDG investment is excluded. To fill in this gap, the [10]–[14] have combined the planning of RDG and network components under the same framework.

Through the above review, it is shown that considerable work has been undertaken concerning methods for improving planning coordination between RDG and network; however, most of them represent the load demand by deterministic multi-state snapshots yet the time-varying characteristics are barely considered. In fact, for real-life practices, another aspect that affects the environmental contribution of RDG comes from the correlation between the renewable energy supply and electricity usage in the *time domain*. As the energy balance in distribution systems must be perfectly satisfied at any time, DISCO could always have to purchase carbon-intensive electricity from the main grid to serve the demand and curtails RDG in other times, even when the primary energy supply is high. The mismatch in the temporal dimension would however pose an adverse impact on the actual emission benefits created by RDG.

To address this issue, one feasible solution is to introduce time-variant demand response (DR), such as real-time pricing (RTP) mechanism [15], [16]. As RTP offers a fiscal tool to reshape the demand consumption pattern for more closely follows the availability of renewable energies, once the penetration of DR reaches a certain threshold, its effect for promoting efficient exploitation of RDG can be remarkable [17]. Nevertheless, activating DR implies the investment of smart metering (SM) and other infrastructures that enable two-way communication between the distribution system operator (DSO) and end-users. Regarding the diversity and ambiguity of consumer behaviors, there is a concern about whether the potential benefits from energy demand management of RTP could outweigh the required investment costs or not [18].

From the above explanation, it can be seen that decarbonization in distribution systems heavily depends on the efficient utilization of RDG, which is both related to the network capacity and system loading level. Also, the optimal distribution of DR infrastructures is affected by the controllability of demand and connection with RDG. As such, an integrated planning methodology for a transition to low-carbon distribution systems is proposed in this study, which is hereinafter referred to as low-carbon planning (LCP). It is called integrated, because different types of DG (including renewable and non-renewable) and SM are considered as equivalent resources and optimized with the network investment simultaneously. The problem is formulated as a dynamic model with two inter-related stages to minimize the sum of economic and CO₂ emission costs over planning horizons. For more realistic applications, the uncertainties associated with renewable energy resources and demand responsiveness are also taken into account by using a probabilistic scenario-based approach.

The main contributions of this study are summarized as follows:

- 1) A methodology which for the first time applies SM as a kind of optional planning resource to promote CO₂ abatement at the distribution level is proposed.

- 2) To suit the features of LCP, a nodal-based DR representation is developed, wherein the fading effect during the load recovery of DR activities is highlighted.
- 3) Valuation of DR benefits to the system is handled by using a scenario-based RTP offering model, which effectively captures the time-varying uncertainties of wind generation and customer behaviors.

It is worth mentioning that the methodology proposed in this paper is related to a central planning context, where distribution systems are both owned and operated by specialized DISCOs. However, the framework of LCP can be also applicable to the liberalized electricity markets with extensions to incorporate other relevant entities (like DG operators) and merge their decision-makings in the model formulation without difficulty.

The rest of this paper is organized as follows. Section II gives an overview of the proposed LCP paradigm. Representations of system generation and price-responsive demand are elaborated in Section III. Subsequently, the detailed formulation of LCP is presented in Section IV, and the introduction of optimization procedures is given in Section V, which forms the core part of this work. In Section VI, the application of LCP in the IEEE-33 system is analyzed. Finally, the conclusions are drawn in Section VII.

II. FRAMEWORK OF THE PROPOSED LCP

Overall, the carbon footprint of distribution systems entails two aspects: the direct generation emissions by the purchased energy and indirect emissions embodied in the asset material of system components [19]. This implies that planning for CO₂ emission minimization needs to consider all the possible system operating states, instead of merely focusing on the extreme conditions as the traditional network planning does.

As such, the proposed LCP model is formulated by two stages, corresponding to the decision-making in the planning and operation phases respectively. As shown in Fig. 1, the first-stage involves network reinforcement and planning of RDG and DR resources. Decision variables include the optimal timing, locations and capacities of feeders, DG units and SM to be upgraded or installed. All the candidate planning proposals will then be transferred to the second-stage as *a priori*. In its place, DSO combines network data, forecasted generation-demand data, as well as day-ahead market prices to determine the optimal variation of offering prices, and transmits above information to the customers with SM installed. The prices are designed to encourage DR to behave such that the interests of DSO could be maximized. Meanwhile, the feasibility of first-stage decisions is also checked in these simulations, where the technically infeasible solutions are identified and discarded. After this step, the posted DR load will be fed back to revise previous planning scheme in the first stage. With on-going simulations, these procedures would finally arrive at the optimal solution.

III. REPRESENTATION OF SYSTEM GENERATION AND PRICE-RESPONSIVE DEMAND

In this work, we have considered two types of renewable generation, i.e., wind-based DG (DWG) and photovoltaic (PV), and gas turbines (GT) as a conventional firm generation in the

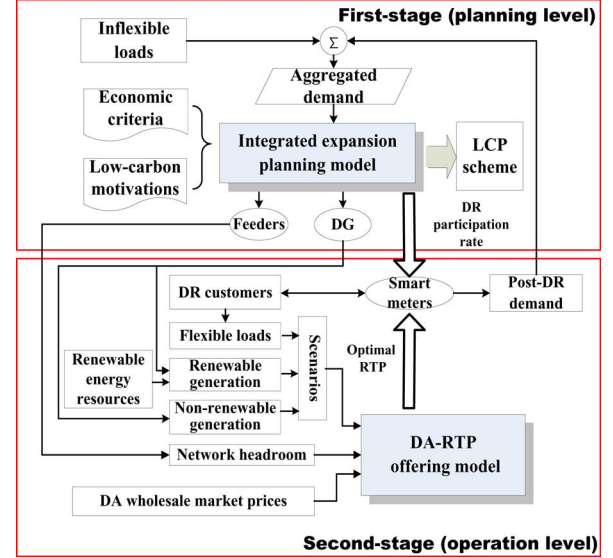


Fig. 1. Framework of the proposed LCP.

model, in view of their promising prospects in China. However, other types of RDG (such as tidal power, small hydro, etc.) may also be incorporated in the model.

A. Wind Generation Modeling

The generation output of wind turbines mainly depends on the wind speed at the site. From a long-term perspective, a Weibull distribution [4], [11], [12] is most commonly used to describe the wind variability. Then, wind power output is assumed following a piecewise function [4], [11] of the wind velocity with given technical parameters, e.g., cut-in, cut-off, rated speeds, and rated power, of the DWG units.

B. PV Generation Modeling

Numerous studies have demonstrated that the stochastic solar radiation can be properly represented by a Beta distribution [4], [20]. The relationship between radiation intensity and the output power of a PV module is described by a linear function, which is presented in [20].

In this study, it is also assumed that both wind turbines and PV are operated at the unity power factor [4], [11].

C. GT Generation Modeling

Due to the unpredictability of RDG, a certain percentage of dispatchable DG is still needed so as to provide ancillary services such as spinning reserves. In this work, GT is taken into account as a type of fast-response generation, the output of which could maintain constant at any requested value within the operating limits without uncertainties.

D. RTP-Responsive Demand Modeling

If the SM penetration in customer groups is denoted by κ , then the nodal demand $D(t)$ satisfies the relation, $D(t) = (1 - \kappa)D(t) + \kappa D(t)$. The first and second terms represent the aggregated inflexible and flexible loads, which is substituted by $D^{ure}(t)$ and $D_0^re(t)$ hereinafter. Since only flexible loads can be influenced by price variations, $D_0^re(t)$ will be further analyzed.

1) *Real-Time DR*: With the concept of elasticity [21], [22], the customer response to RTP as single-period modified demands are expressed as [22]

$$\tilde{D}^{re}(t) = D_0^{re}(t) \left\{ 1 + \frac{\varepsilon(t)[\rho(t) - \rho_0(t)]}{\rho_0(t)} \right\}. \quad (1)$$

As can be seen, the expected load variation due to DR is related to the initial forecasted load, electricity tariff before and after imposing RTP, as well as the price-responsiveness of customers in corresponding periods.

2) *Load Recovery With the Fading Effect*: The decrease in electricity usage would be detrimental to utility profits. To compensate for the DR-imposed losses and maintain the initial balance, customers are prone to consciously increase the consumption in other time periods. This is often referred to as “load recovery” [23], [24].

In [23], the phenomenon is modeled by a demand payback, which is evenly distributed in the time slots upon the reduction. However, due to the natural property of appliances, the relative satisfaction of completing a certain consumption activity is at its peak when it is originally settled, and diminishes over time as prolonging waiting time increases the inconvenience to the consumers [25]. This makes the distribution of corresponding recovered demand commensurate with temporal adjacency to the reduction event. Because of this, for accurate simulations, the above fading effect (FE) must be considered in DR modeling.

Varying to the control strategy adopted, the total recovery energy in DR is not necessarily equal to can be greater than the earlier reduction [25]. As such, let us use an additional parameter ϑ to denote the ratio of payback to reduced consumption, typically $\vartheta \geq 1$, then the load restoration with respect to the reduction event in period t can be indicated as

$$\sum_{t' \in TR} P^{\text{cov}}(t, t') r_t = [D_0^{re}(t) - \tilde{D}^{re}(t)] r_t \vartheta. \quad (2)$$

Here, $P^{\text{cov}}(t, t')$ stands for the amount of paid-back consumption in the recovery interval t' . Owing to the utility decay, it is not constant, but rather should be seen as a time variable. Without loss of generality, the demand here is assumed to recover linearly with time at a specific rate of ϖ , which indicates the degree of diminishing tendency. As a rational DR customer may not only choose to defer but can also pre-schedule the electricity usage upon pricing signals [24], the payback demand for recovery interval TR is thus derived in a piecewise equation with a symmetric hypothesis as

$$P^{\text{cov}}(t, t') = \begin{cases} [D_0^{re}(t) - \tilde{D}^{re}(t)]\vartheta + \varpi \cdot (t - t') & t' < t \\ [D_0^{re}(t) - \tilde{D}^{re}(t)]\vartheta - \varpi \cdot (t' - t) & t' > t \end{cases} \quad \forall t' \in TR \quad (3)$$

An illustration of load recovery with the fading effect is given in Fig. 2.

3) *Aggregated Load Demand*: By combining the expected load reduction (1) and accumulated payback demand (2)

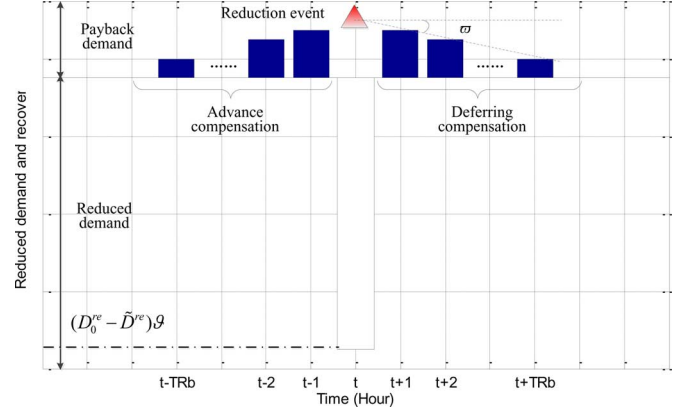


Fig. 2. Load recovery with the fading effect.

throughout the recovery session, we will have the final aggregated demand as

$$D(t) = D^{ure}(t) + \tilde{D}^{re}(t) + \sum_{t' \in TR} P^{\text{cov}}(t', t). \quad (4)$$

The outcome from (4) is the posted demand under the idealized situation. However, customer compliances to RTP may vary considerably in reality, due to diversity in rationales, appliance composition and other unpredictable factors. It is difficult to obtain the accurate individual demand curve of all the users; besides, whether customers are willing to disclose their privacy in terms of electricity consumption also remains questionable. Such uncertainties make the parameters $\varepsilon_i(t)$, ϑ , and ϖ highly stochastic.

For distribution systems containing a large number of independent loads, particularly at the medium voltage level, the aggregated form of DR whereas exhibits statistical regularity. Some advanced forecasting techniques (like [26]) could be used to estimate the associated probability density function of price-responsiveness. In [27], a Gaussian distribution is applied to represent elasticity variations. Considering that elasticity should also yield to limits on the maximum and minimum credible values, a truncated Gaussian distribution is used in this work:

$$f^d(\varepsilon(t)) = \begin{cases} 0 & \varepsilon(t) < \varepsilon_{\min}(t) \\ \frac{1}{\sqrt{2\pi}\sigma^d(t)} e^{-\frac{(\varepsilon(t) - \mu^d(t))^2}{2(\sigma^d(t))^2}} & \varepsilon_{\min}(t) \leq \varepsilon(t) \leq \varepsilon_{\max}(t) \\ 0 & \varepsilon(t) > \varepsilon_{\max}(t) \end{cases} \quad (5)$$

where $\mu^d(t)$ and $\sigma^d(t)$ stand for the statistical mean and standard deviation of the load in the specific time period t .

ϑ and ϖ are assumed to be evenly distributed within the range $[1, \vartheta_{\max}]$ and $(0, \varpi_{\max}]$, where $\vartheta_{\max}$ and $\varpi_{\max}$ are the upper limits of the deviation of ϑ and ϖ , respectively.

E. Generation of Synthetic Scenarios

The aforementioned uncertainties can be incorporated into the planning model by means of synthesizing into a set of scenarios [4], [12]. Each scenario represents a possible operating state of the system in the planning horizon. If we assume the

variations of uncertain factors independent of each other, the probability of occurrence assigned to scenario s yields

$$PR_s = \underbrace{\int_{\mathbf{x}_{NU,l,s}}^{\mathbf{x}_{NU,u,s}} \cdots \int_{\mathbf{x}_{1,l,s}}^{\mathbf{x}_{1,u,s}}}_{NU\text{-dimension integral}} \left[\prod_{i=1}^{NU} f_i(\mathbf{x}_i) \right] d\mathbf{x}_1 \cdots d\mathbf{x}_{NU}. \quad (6)$$

In the above equation, $\mathbf{x}_i$ corresponds to wind speed, solar radiation, elasticity, and recovery ratio, thus $NU = 4$. By adjusting the upper and lower bounds (with subscripts u and l), planners may define the number of scenarios according to their preferred trade-off between simulation precision and computation burden.

IV. PROBLEM FORMULATION FOR LCP

A. Objective Function

As the metric of CO₂ emissions can be converted into the monetary form by carbon taxation, the objective function (OF) of LCP equates the minimization of the total economic and CO₂ emission costs over planning horizons, which can be briefly expressed as

$$\text{OF} = A + B \quad (7)$$

where A is the net present value (NPV) of system investment costs, which are derived from the first-stage planning decision; B stands for the NPV of total expected variable costs under all the credible system operating states, which is the recourse sub-OF for the second stage model.

The details of A are given as follows:

$$\begin{aligned} A = & \sum_{\lambda=1}^{TY} \frac{1}{(1+d)^\lambda} \left[\sum_{(i,j) \in \Omega_F} C_a^{fdr} l_{ij} \tau_{ij,a,\lambda} + \sum_{g \in \Omega_G^{gt}} C_g^{gt} n_{g,\lambda}^{gt} \right. \\ & + \sum_{g \in \Omega_G^w} C_g^w n_{g,\lambda}^w + \sum_{g \in \Omega_G^{pv}} C_g^{pv} n_{g,\lambda}^{pv} \\ & + \sum_{k \in \Omega_D} C_k^{sm} (\kappa_{k,\lambda}^{sm} - \kappa_{k,\lambda-1}^{sm}) n_k^{hd} \\ & + C_\lambda^{ct} \left(\sum_{(i,j) \in \Omega_F} \sum_{a \in \Omega_a} e b_a^{fdr} l_{ij} \tau_{ij,a,\lambda} + \sum_{g \in \Omega_G^{gt}} e b_g^{gt} n_{g,\lambda}^{gt} \right. \\ & + \sum_{g \in \Omega_G^w} e b_g^w n_{g,\lambda}^w + \sum_{g \in \Omega_G^{pv}} e b_g^{pv} n_{g,\lambda}^{pv} \left. \right) \left. \right] \\ & - \frac{1}{(1+d)^{TY}} \left(\sum_{(i,j) \in \Omega_F} Z_a^{fdr} + \sum_{g \in \Omega_G^{gt}} Z_g^{gt} \right. \\ & + \sum_{g \in \Omega_G^w} Z_g^w + \sum_{g \in \Omega_G^{pv}} Z_g^{pv} + \sum_{k \in \Omega_D} Z_k^{sm} \left. \right). \quad (8) \end{aligned}$$

In (8), the first three lines represent the capital cost of feeders, GT, DWG, PV, and SM, respectively. As the CO₂ content in SMs is relatively negligible, it is not taken into account in the calculation of embodied emission tax (the fourth and fifth lines).

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CO₂ content is herein represented by the carbon footprint induced during the manufacturing phase and deemed as equivalent regardless of their timing [19].

Using NPV as the OF of the planning model implies that the horizon for planning implementation should be consistent with the lifespan of the candidate resources used. To comply with this intrinsic requirement, the residual value Z of each equipment is also considered in (8), which is expanded as

$$Z = \frac{C}{TL} [TL - (TY - \lambda)]. \quad (9)$$

The outcome of (9) gives the residual value of equipment at the end of the planning horizon once their capital costs C , lifespan TL , and investment year λ are known. It needs to be deducted in the OF. In this way, the equipment with different lifespans could be used as equal resources under the same planning horizon.

As the second-stage represents tariff optimization through repeated simulations for minimizing the expected operating costs, B in (7) (second-stage OF) includes the expense for purchasing power from the grid, GT production cost, network loss cost as well as the corresponding carbon tax created, and is given by

$$\begin{aligned} B = & \sum_{\lambda=1}^{TY} \frac{1}{(1+d)^\lambda} \sum_{s=1}^{NS} PR_s \\ & \times \sum_{t=1}^{TH} r_t \left\{ \rho_\lambda^{gsp}(t) P_s^{gsp}(t) \right. \\ & + \rho^{gt} \sum_{g \in \Omega_G^{gt}} P_{g,s}^{gt}(t) + \rho^{los} P_s^{los}(t) \\ & + C_\lambda^{ct} \left[e g_\lambda^{gsp} P_s^{gsp}(t) + e g^{gt} \sum_{g \in \Omega_G^{gt}} P_{g,s}^{gt}(t) \right] \left. \right\}. \quad (10) \end{aligned}$$

In (10), the total variable cost for each scenario is simply calculated by multiplication of the energy obtained from non-renewable power sources (including the main grid and GT), its related generation emissions, and network losses by the market acquisition price ρ^{gsp} (production cost ρ^{gt} for GT), emission tax rate C^{ct} , or loss cost ρ^{los} respectively, and summed together over the periods from $t = 1$ to $t = TH$ comprising one year. Here, P_s^{gsp} , P_g^{gt} and P^{los} are dependent variables, which are essentially the functions of the posted price vector ρ_s in scenario s .

B. Constraints

As the first-stage model is associated with the general (not necessarily technical feasible) planning schemes, it is subjected to the following two constraints.

1) *SM Saturation Constraint*: Obviously, the volume of SM is limited by the household number in each load point as

$$0 \leq n_{k,\lambda}^{sm} \leq n_k^{hd} \quad \forall k \in \Omega_D, \forall \lambda \in TY. \quad (11)$$

2) *Unicity of Feeder Type*: Only one type of feeder conductor is allowed to be used in each single right-of-way, as is assured by

$$\sum_{a \in \Omega_a} \tau_{ij,a,\lambda} = 0 \text{ or } 1 \quad \forall (i, j) \in \Omega_F, \forall \lambda \in TY. \quad (12)$$

For the second-stage model, in order to obtain the optimal pricing variations on the premise of a technical feasible planning scheme, a variety of constraints should be satisfied for any operating scenario s as follows:

$$\begin{aligned} & P_{i,s}^{\text{gsp}} + P_{i,s}^{\text{gt}} + P_{i,s}^{\text{pv}} + P_{i,s}^w - P_{i,s}^{\text{cur}} - D_{i,s} \\ &= \sum_{j \in \Omega, j \neq i} V_{i,s} V_{j,s} Y_{ij} \\ & \quad \times \cos(\theta_{ij} + \delta_{j,s} - \delta_{i,s}) \quad \forall i \in \Omega \end{aligned} \quad (13)$$

$$\begin{aligned} & Q_{i,s}^{\text{gsp}} + Q_{i,s}^{\text{gt}} - Dq_{i,s} \\ &= - \sum_{j \in \Omega, j \neq i} V_{i,s} V_{j,s} Y_{ij} \sin(\theta_{ij} + \delta_{j,s} - \delta_{i,s}) \quad \forall i \in \Omega \end{aligned} \quad (14)$$

$$V_{i,\min} \leq V_{i,s} \leq V_{i,\max} \quad \forall i \in \Omega \quad (15)$$

$$0 \leq I_{i,j,s} \leq I_{i,j,\max} \quad \forall (i, j) \in \Omega_F \quad (16)$$

$$0 \leq P_s^{\text{gsp}} \leq \text{Cap}_{\max}^{\text{sb}} \quad (17)$$

$$|P_s^{\text{gsp}}(t) - P_s^{\text{gsp}}(t-1)| / r_t \leq \psi_{\max} \quad (18)$$

$$1 - \left[\sum_{g \in \Omega_G^{\text{rdg}}} P_{g,s}^{\text{cur}} / \left(\sum_{g \in \Omega_G^{\text{pv}}} P_{g,s}^{\text{pv}} + \sum_{g \in \Omega_G^w} P_{g,s}^w \right) \right] \geq \alpha_{\text{tol}} \quad (19)$$

$$\zeta_{k,\min} \leq \tilde{D}_{k,s}^{\text{re}} / D_{0,k,s}^{\text{re}} \leq \zeta_{k,\max} \quad \forall k \in \Omega_D \quad (20)$$

$$\rho_{k,\lambda}^{\text{gsp}} \leq \rho_{k,s,\lambda} \leq \rho_{k,\lambda,\max} \quad \forall k \in \Omega_D, \quad \forall \lambda \in TY. \quad (21)$$

The conventional equality constraint for the system active/reactive power flow is represented by (13) and (14), respectively. Equations (15) and (16) assure the magnitude of nodal voltage and current in all the feeders kept within their permitted limits during operations. The power input from the external grid must be limited below the substation capacity $\text{Cap}_{\max}^{\text{sb}}$, and reverse power flow is not allowed, which is assured by (17). Seen from the distribution side, external grid could be equivalent to a conventional generating unit with a large capacity. Likewise, the constraint on the ramping rate likewise is also applicable to the external power supply, in order to represent the limited supply from the main grid and maintain its stable operations. This is enforced by (18). Equation (19) confirms the utilization factor of RDG (defined as the ratio of actual energy output to their potential) not lower than the minimum reasonable level that is tolerable α_{tol} . As the demand of DR customers is not totally influenced by prices, limits concerning the DR potential should also be considered by (20). Furthermore, RTP may expose customers to high risks in the electricity bill. To hedge against unbearable volatility directly transferred to the customers [22], the tariff is thus considered being higher than the market acquisition price, but not over the “cap” predefined, which is represented by (21).

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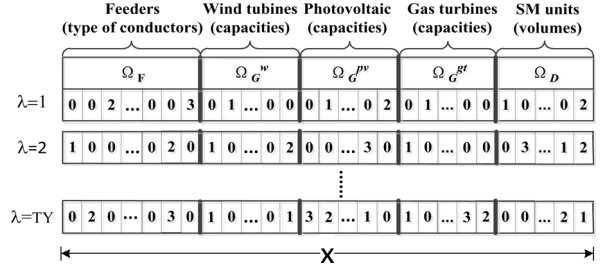


Fig. 3. Structure of coded candidate solutions.

V. SOLUTION METHODOLOGY

Mathematically, the above LCP model is a large-scale mixed-integer nonlinear programming (MINLP) problem, which cannot be properly handled by the conventional optimization techniques. Hence, evolution-based heuristic searching methods, such as genetic algorithm (GA), have been widely accepted as an efficient tool to deal with MINLP [13]. In this study, an interior-point-method-embedded discrete GA (IPM-DGA) [28] is employed so as to solve the model with high performance and accuracy.

For this hybrid algorithm, DGA is firstly dedicated to find candidate planning schemes by rounding off decision variables for the first-stage sub-problem. As the planning decision in each year is made based on the existing network of the previous years, DGA is customized with integer codification and applied to the problem. Each solution is represented by a $TY \times x$ matrix corresponding to TY planning years and x gene groups, as is shown by Fig. 3. The rows of the matrix are arranged to be token feeders, DWG, PV, GT and SM units with assigned values indicating their capacities (volumes) to be upgraded or installed. The length of x depends on the number of right-of-ways and candidate installation points comprised in the system.

Subsequently, IPM is applied to determine the optimal real-time tariff offered to DR customers. During simulations, the modified system demand under DR is compared with the total available power from RDG. If there is a surplus in supply, the extra part would be curtailed from RDG, $P^{\text{cur}} = P^{\text{rdg}} + P^{\text{gsp}} - D$. Here, the power import from the grid must still be included as it is limited by the “ramping down” constraint of energy transactions. If a deficit occurs, the available power from the external grid is used first. Only if the load still cannot be served, GT needs to be dispatched finally. Following this principle, the operation cost of candidate solutions (i.e., B in (7)) can be determined.

The fitness function used for optimization is a composite index, which is composed of the OF in (7) with a penalty factor, as is given in the following:

$$\text{Fitness}(k) = \text{OF} + \text{PF} \cdot \sum_{c=1}^{nc} \omega_c \quad (22)$$

where PF is a large number (taken as 10^8 here); nc the total number of model constraints. If any constraint violation in the load flow calculation is identified, the corresponding penalty factor ω_c is set to 1 so as to enable the “death penalty”. Ranking

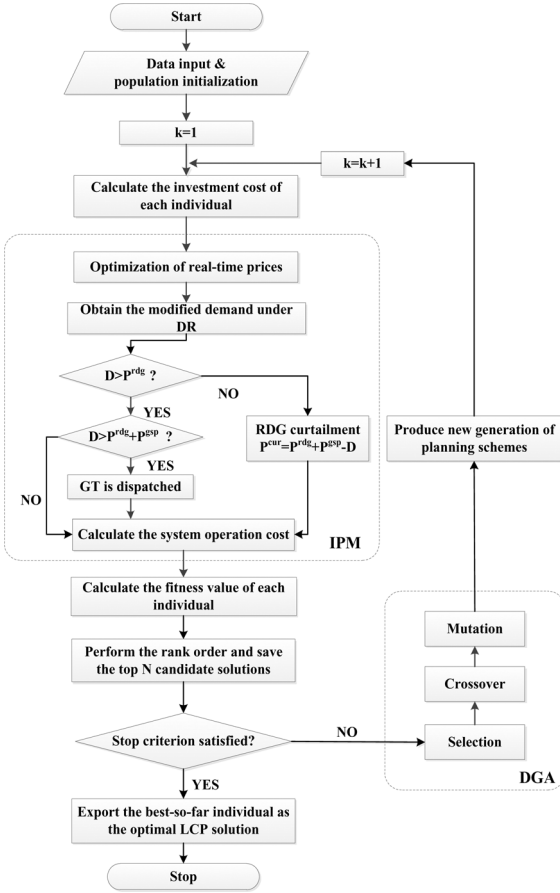


Fig. 4. Flowchart of the hybrid algorithm.

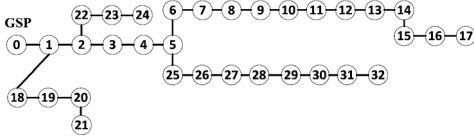


Fig. 5. 32-bus distribution system test case.

of the candidate solutions according to the fitness values of individuals is performed and the set of superior plans is updated in each round.

The optimization procedure stops and exports the final results if either of the following two termination criteria is satisfied:

- 1) $Fitness(k-1) - Fitness(k) < \beta$, where β is the pre-specified tolerance indicating the accuracy of convergence; or
- 2) $k = N$, the maximum number of iterations N is reached.

Fig. 4. illustrates the flowchart of the above hybrid algorithm.

VI. CASE STUDY

A. Test System and Basic Data

To assess the effectiveness of LCP, a 32-bus radial distribution system shown in Fig. 5 is chosen for the case study [29]. The system contains a mix of residential, commercial, and industrial customers being connected to the main grid via a HV/MV substation with the rating of 15 MVA at Bus-0.

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TABLE I
DATA OF AVAILABLE PLANNING RESOURCES

Option		Capital cost	eb	eg (kg/kWh)	TL (year)
Feeder	LGJ-35	\$6638/km	6.7 t/km	N/A	15
	LGJ-70	\$8427/km	14.1 t/km	N/A	15
	LGI-120	\$10930/km	23.6 t/km	N/A	15
DWG		\$1227/kW	32.1 t/unit *	0	20
PV		\$2100/kW	9.5 t/unit *	0	25
GT		\$825/kW	5.3 t/unit *	0.77	20
SM		\$120/ unit	N/A	N/A	20

* Referred values for PV/ DWG/ GT units with the capacity of 0.1/0.2/0.3 MW [33]–[35]

TABLE II
TEST CASES

Case	#1	#2	#3	#4	#5	#6	#7
Network reinforcement	✓	✓	✓	✓	✓	✓	✓
DR	×	×	×	×	✓	✓	✓
DG type	DWG	×	✓	×	✓	×	✓
	PV	×	×	✓	✓	×	✓
	GT	×	✓	✓	✓	✓	✓

TABLE III
COMPARISON OF RESULTS FOR DIFFERENT PLANNING PARADIGMS

Case		#1	#2	#3	#4	#5	#6	#7
Cap. costs (M \$) (res. value incl.)	FD	0.36	0.37	0.37	0.36	0.36	0.36	0.35
	DWG	N/A	2.30	N/A	1.49	2.30	N/A	1.54
	PV	N/A	N/A	3.36	1.03	N/A	3.46	1.03
	GT	N/A	0.31	0.05	0.19	0.19	0	0.06
	SM	N/A	N/A	N/A	N/A	0.05	0.04	0.04
	ECT	0.01	0.01	0.02	0.01	0.01	0.01	0.01
Oper. costs (M \$)	EPC	8.96	6.34	6.10	5.75	6.02	5.97	5.35
	NLC	0.91	0.62	0.58	0.46	0.50	0.49	0.41
	GGC	-	0.34	0.15	0.25	0.20	0	0.04
	GCT	2.32	1.67	1.57	1.43	1.55	1.52	1.35
Total cost (M \$)		12.56	11.96	12.20	10.97	11.18	11.85	10.18
Cost savings (%)		-	4.78	2.87	12.66	10.98	5.65	18.95
DG gen. (10 ⁵ MWh)	RDG	-	1.02	1.19	1.39	1.23	1.27	1.54
	GT	-	0.07	0.01	0.06	0.04	0	0.01
RDGP (%)		-	24.46	30.38	34.23	30.01	31.96	37.05
CO ₂ emissions (10 ⁵ t)		3.28	2.33	2.18	1.99	2.15	2.11	1.86
ERR (%)		-	28.96	33.54	39.33	34.45	35.67	43.29

*FD: feeders; ECT: carbon tax of embodied emissions; EPC: energy purchase cost; NLC: network loss cost; GGC: GT generation cost; GCT: carbon tax of generation emissions; RDGP: RDG energy as a percentage of total system consumption; ERR: emission reduction rate.

To determine the effective locations for SM and DG placement, a preliminary sensitivity analysis is taken. The impact factor of the allocation at all the system buses is calculated and ranked in a descending order according to their sensitivity to the objective function value of the system. Then, the candidate locations are chosen from top positions with respect to each type of equipment, as $DWG \in \{3, 17, 21\}$; $PV \in \{16, 19, 30, 31\}$; $GT \in \{2, 10\}$; $SM \in \{2, 3, 7, 12, 13, 18, 19, 23, 24, 27, 28, 30\}$. Here, the number of RDG nodes which are taken into account is determined arbitrarily based on the sensitivity analysis (which will be elaborated on Section VII-A), although more nodes could

TABLE IV
OPTIMAL PLANNING SCHEMES OF DIFFERENT PARADIGMS

Stage	Resource	#2	#3	#4	#5	#6	#7
I	DWG(MW)	0.6 at bus 3; 0.4 at bus 17; 0.6 at bus 21	N/A	0.2 at bus 3; 0.6 at bus 17; 0.4 at bus 21	0.6 at bus 3; 0.4 at bus 17; 0.6 at bus 21	N/A	0.4 at bus 3; 0.6 at bus 17; 0.2 at bus 21
	PV(MW)	N/A	0.4 at bus 16, 30; 0.3 at bus 31	0.2 at bus 16, 31	N/A	0.4 at bus 16, 30; 0.3 at bus 31	0.2 at bus 16, 31
	GT(MW)	0	0	0	0	0	0
	SM(%)	N/A	N/A	N/A	10 at bus 13; 30 at bus 2,7,28; 40 at bus 24; 50 at bus 19,30;	10 at bus 19,23,28; 20 at bus 7,13,18,24,30; 30 at bus 2,12;	10 at bus 13,28; 30 at bus 2,7,12,24,30; 40 at bus 18;
II	DWG(MW)	0.6 at bus 3;	N/A	0.2 at bus 21	0.6 at bus 3;	N/A	0.2 at bus 21
	PV(MW)	N/A	0.5 at bus 30, 31	0.2 at bus 30	N/A	0.1 at bus 16; 0.5 at bus 30, 31	0.2 at bus 30
	GT(MW)	0.6 at bus 2	0	0.3 at bus 2	0.3 at bus 2	0	0
	SM(%)	N/A	N/A	N/A	20 at bus 3, 7; 30 at bus 13, 28; 40 at bus 12,24,18 50 at bus 2, 23, 27	10 at bus 7, 12,24; 20 at bus 3, 23,27,30; 30 at bus 13, 19,28;	10 at bus 7, 24; 20 at bus 3,12,18,23,27; 30 at bus 13,23,28; 50 at bus 30
III	DWG(MW)	0.4 at bus 17;	N/A	0	0.2 at bus 17; 0.2 at bus 21;	N/A	0.2 at bus 17
	PV(MW)	N/A	0.7 at bus 19	0.2 at bus 19	N/A	0.7 at bus 19	0.2 at bus 19
	GT(MW)	0.3 at bus 10	0.3 at bus 2	0.3 at bus 10	0.3 at bus 10	0	0.3 at bus 10
	SM(%)	N/A	N/A	N/A	10 at bus 23; 20 at bus 12, 13; 30 at bus 18, 19, 30	10 at bus 23,30; 20 at bus 12,18; 40 at bus 2	10 at bus 23; 20 at bus 12,18,24,30; 30 at bus 2, 19, 27

be used out of the interests of the planner. For real applications, it should be noted that many other factors must also be considered in the selection of RDG candidate locations, such as geographical condition, landscape aesthetics, etc.

The customer load curves are derived from the real historical data in Beijing. Demand elasticity extracted from [21] is used for each planning year. The reference fixed tariff for industrial residential, and commercial users is 5.2, 6.9, and 7.3 cents/kWh [10]. For simplicity, it is assumed that there are 100 consumers of the same type at each bus. Besides, the load recovery session over the previous and subsequent of 4 h [30] is considered.

The planning is implemented in the time horizon of 15 years with intervals every 5 years, which is consistent with the engineering recommendation of China Distribution System Planning Directive [31]. The annual load growth rate of 2% and discount rate of 8% is assumed. The data of planning resources is given in Table I [10], [19], [32]–[35]. The capacities of DG are discretized at a definite step of 0.1(PV), 0.2 (DWG) and 0.3 (GT) MW, according to their typical commercial sizes. We have the GT generation cost $\rho^{\text{gt}} = 8.5$ cents/kWh and network loss cost $\rho^{\text{los}} = 3.6$ cents/kWh. Voltage limit is set to $\pm 5\%$ of the nominal value, the minimum RDG utilization of 90% of their installed capacity, and the price cap below 150% of the base fixed price $\rho_0(t)$. Moreover, the carbon tax rate and grid emission intensity is initially taken as \$10/t and 0.92 kg CO₂/kWh [10], respectively.

For comparison purposes, seven different cases are studied as shown in Table II, where Case-1 and Case-7 represents the base case (as presented in [19]) and the proposed LCP, respectively.

According to the experiences of repeated trials, the proposed IPM-DGA is executed with the following parameters: population size = 50; crossover rate = 0.60; mutation rate = 0.05; max-

imum iterations = 250. The simulation is programmed and implemented in the MATLAB environment.

B. Simulation Results

Table III gives the optimization results of the study, with planning details of DG and SM tabulated in Table IV.

As is revealed, no matter DR is included or not, there is a significant reduction of total cost and CO₂ emissions with the addition of RDG, as compared to the benchmark.

Although both wind and solar resources involve significant uncertainties, the environmental contribution of wind generation to system emission benefit (Case-2) is found weaker than the solar (Case-3) in terms of total energy generation. This can be possibly explained from the difference in their production characteristics. The wind potential is generally at the peak during the night or early morning whilst the load is relatively low. Such a mismatch prevents the efficient operation of DWG to its full potential. Moreover, to avoid abrupt interruptions of RDG, a higher degree of backup power from GT is also needed in the wind case. This implies the supply of solar power may be more consistent with the load profile in practices. However, the high cost in PV investment still poses a significantly negative effect on the overall attractiveness of the solar case. In spite of the above differences, due to the singleness of the primary resource, it can be seen that the contribution of RDG is limited in the total energy supply for both cases above, which takes up 24.46 and 30.38%, respectively.

In Case-4, the simultaneous planning of DWG, PV, and GT is performed. As indicated in Table III, the coordination between multiple types of DG facilitates the utilization of more RDG energy by end-users overall and minimizes the input of carbon-intensive electricity from GT or the main grid, hence leading to

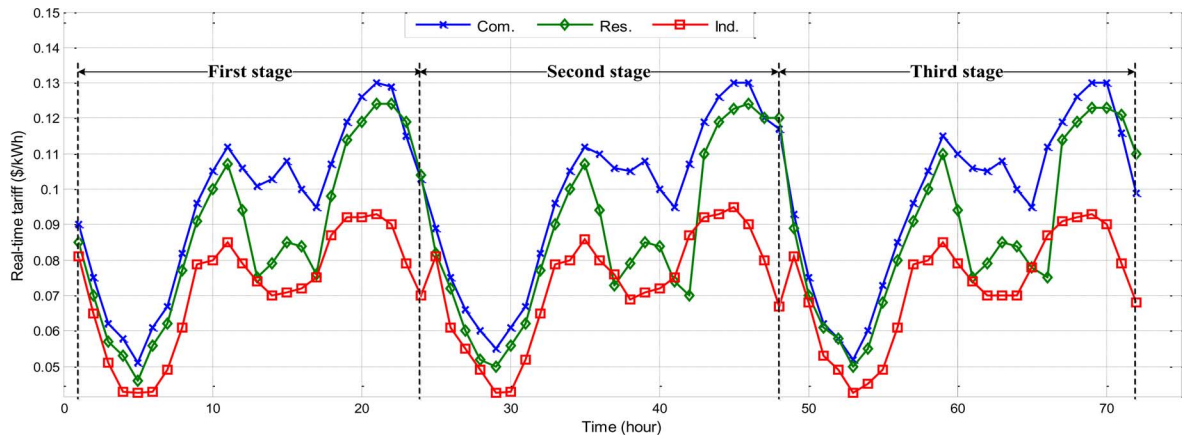


Fig. 6. Real-time tariff offered to different types of customers in Case-7.

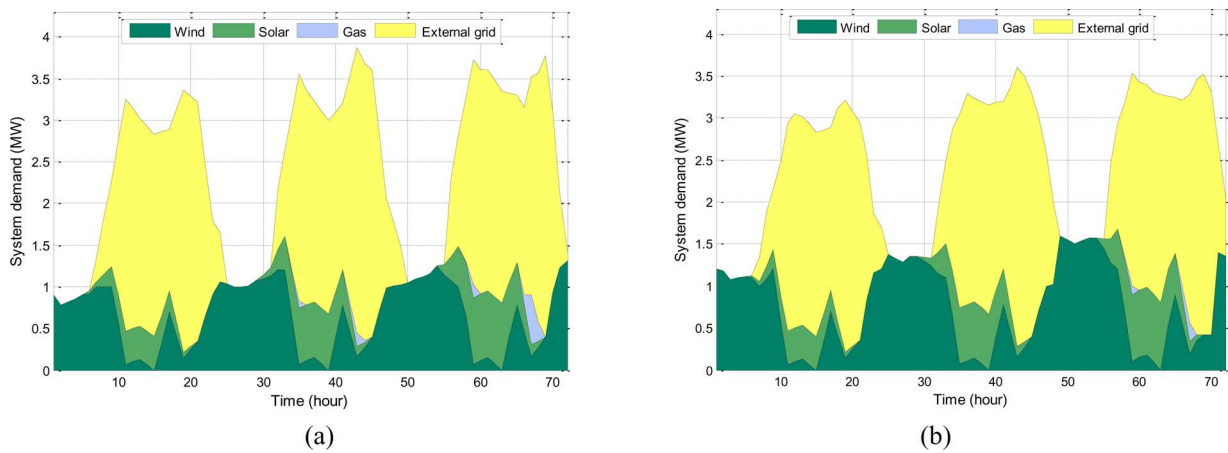


Fig. 7. Demand consumption with breakdown of energy sources in the typical day of Cases-4 (a) and 7 (b).

a lower operation cost and CO_2 emissions. Meanwhile, diversified investment leads to smaller capacity allocated for each type of DG, which is helpful in decreasing the curtailment risk. As a consequence, there is a rise in the total energy import from RDG. This demonstrates that using wind and solar simultaneously is more beneficial for the continuous production of RDG than using them separately due to the inherent complementarity between both resources and can bring about more preferable CO_2 abatement.

For Cases 5 to 7, the option of DR is considered. As can be seen, there is a further improvement of economic and environmental benefits achieved in these cases as compared to the corresponding situation without DR. The scheme is found to be optimal when the wind, solar, gas, and SM resources are taken into account simultaneously, with the total expected cost of \$10.18 M and CO_2 emissions of 186 000 tons.

In order to reveal the effect of DR to such results, one day is selected arbitrarily from each stage (the same date is used here), and variation of the optimized real-time tariff in Case-7 is shown in Fig. 6.

It is seen that the prices offered to customers vary widely in different times of the day under RTP. The highest values generally appear in the periods of 6 to 9 p.m. (when the wind is low while the solar output is weakened due to sunset), whereas the lowest ones are concentrated between 2 to 6 a.m. (when the wind resource is at peak). A further comparison is made on the

load patterns between Cases-4 and 7, along with the breakdown of their energy sources, as shown in Fig. 7.

It is clear that enabling demand flexibility indeed smooths out the electricity consumptions. Consequently, it gives rise to an obvious higher utilization of RDG by both relieving power import during the deficit of renewable supply, and allowing more additional usage of green power when wind/solar radiation remains surplus. This would bring undoubted benefits. Also, as a smooth load pattern of the system makes energy import from the grid less constrained by the transaction rate limit [(18)], the dependence on the firm generation backup can be largely avoided. Accordingly, there is a reduction of GT in both capacity requirement and actual output when DR is permitted.

From the results of Table III, another particular interest is the difference in DR contributions in different cases. It is shown that DR enables the total cost to decrease by \$0.78 M in the wind scenario, the extent of which is more obvious than the solar case (\$0.35 M). This implies that the potential value of DR resources varies depending on what type of system they are applied in. The larger mismatch between RDG and system demand, the more benefits can be expected from DR implementation.

From the above analysis, it is seen that the proposed LCP is effective in improving the overall efficiency of distribution systems and exhibits more encouraging prospect for CO_2 reduction than other conventional planning approaches.

TABLE V
COMPARISON ON THE VALIDATION OF DR MODEL

Scenario	Total cost (M \$)	CO ₂ emissions (10 ⁵ t)
w/o uncertainty & fading effect	9.50	1.72
w/o uncertainty only	10.03	1.81
w/o fading effect only	9.85	1.77
Proposed DR model	10.18	1.86

TABLE VI
EFFECT OF NUMBER OF RDG INSTALLATIONS ON PLANNING RESULTS

Number of candidate nodes		DWG	PV
1	Total capacity (MW)	0.8	0.5
	Total cost (M \$)	11.97	11.60
2	Total capacity (MW)	1.4	0.6
	Total cost (M \$)	11.05	10.95
3	Total capacity (MW)	1.6	0.6
	Total cost (M \$)	10.18	10.87
4	Total capacity (MW)	1.6	0.8
	Total cost (M \$)	10.11	10.18
5	Total capacity (MW)	1.6	0.8
	Total cost (M \$)	9.96	10.04
7	Total capacity (MW)	1.6	0.8
	Total cost (M \$)	9.86	9.87
10	Total capacity (MW)	1.6	0.8
	Total cost (M \$)	9.86	9.86

C. Validation of DR Model

Table V compares the LCP case with the results when ignoring uncertainties and fading effect in DR modeling.

As revealed, the selection of DR model has a direct impact on the evaluation of planning schemes. Higher cost and emissions take place when the above-mentioned factors are accounted for. This implies the existence of these internal properties could partially offset the benefits created by load redistribution and make price control less pronounced in actual practices. Although the quantitative differences between the cases appear relatively limited, it should be noted that this can be significant for other systems. In that way, neglecting the above-mentioned factors might overestimate the value of DR and lead to suboptimal planning decisions.

VII. DISCUSSION

A. Sensitivity Analysis on the Number of RDG Candidate Buses

As mentioned in Section VI-A, a sensitivity analysis is conducted in our study to investigate the effect of number of RDG installation locations on the planning results. Each type of RDG is considered at up to ten candidate bus locations (as designated in sequence from the top of the list) and the results are presented in Table VI.

It is shown that the total installation of DWG units increases but the system overall cost decreases as more candidate locations are taken into account. Similar conclusions can also be drawn from Table VI with respect to PV. However, after a certain level (as marked in bold), such tendency becomes marginal. The above fact implies that using the particular number of buses for RDG installation (i.e., three for DWG and four for PV) in this study is appropriate. It could give satisfactory solutions and

TABLE VII
CALCULATION OF OBJECTIVE FUNCTION WITH UNCERTAINTIES
HANDLED BY THE PROBABILISTIC AND ARMA MODELS

Length of estimation period for each stage (h)	24	48	72	96	120
ARMA	10.25	10.12	10.09	10.18	10.05
Probabilistic	10.98	9.69	10.40	10.01	10.16
Difference ratio (%)	7.12	4.25	3.07	1.67	1.09

TABLE VIII
OPTIMIZATION RESULTS WITH DIFFERENT CORRELATIONS
AMONG RDG GENERATION AND SYSTEM LOAD

Correlation	Total cost (M \$)	RDGP (%)	CO ₂ emissions (10 ⁵ t)
0.05	12.36	15.25	2.43
0.5	10.89	30.18	1.92
0.95	9.32	58.77	1.60

similar results (in terms of RDG installation and total system cost) with respect to the case with more candidate nodes being considered.

B. Handling of System Uncertainties

In this study, we used a sampling method based on the probabilistic distribution of the underlying uncertainties to generate the scenarios in Section III-E. As a long-term statistical approach, it is simple in concept and particularly suitable for RDG planning studies. However, a disadvantage of this method is the difficulty to capture the autocorrelation of uncertain factors in the data series, which may occur in the wind and solar cases. As such, the calculation of OF is compared with the results using the well-known auto-regressive moving average (ARMA) model wherein the autocorrelation is considered, as shown in Table VII.

The results demonstrate that the precision of the probabilistic model depends on the scale of simulation. In this case, if each stage is represented by a 24-h (one day) estimation (sampling time scale of wind speed and solar radiation is also 24-h), a 7.12% of difference exists between the two models. In contrast, the longer simulation period yields closer outcomes: the difference ratio falls to 1.09% if the estimation is taken on a 120-h basis. This shows that, given an adequate enough sampling scale, the two models lead to close solutions regardless of the consideration of autocorrelation in uncertainties. Also, noted that the fixed distributions are used for the whole period estimation in this work, while the distribution parameters could be updated on a monthly or hourly basis in reality. Hence, the precision of the sampled data by the probabilistic approach could be even improved.

Furthermore, in practice, atmospheric factors may also lead to correlations between RDG generation and system demand. To investigate the impact of this on planning decisions, the wind speed and solar radiation data are simulated [36] with the correlation coefficients of 0.05, 0.5, and 0.95 with respect to the demand, which represents the independent, moderate, and highly correlated situations, respectively. The variation of the optimization results is shown in Table VIII.

As indicated, the solution of LCP is sensitive to the degree of correlations among RDG generation and load. Lower cost and

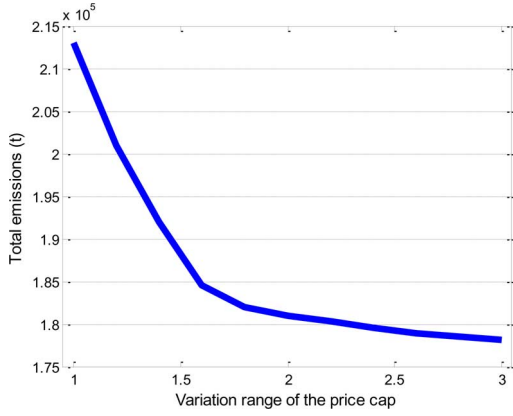


Fig. 8. Impact of the price cap on CO₂ emissions in the system.

emissions, and higher renewable energy utilization are obtained as the correlation factor increases. This means that the effect of LCP essentially varies on different distribution systems. A biased conclusion could be made if such correlations are not considered. Therefore, the credibility of wind/solar historical data is of significant importance in LCP, which should be paid enough attention by planners.

C. Impact of Price Cap on System Emission Reduction

Variation of the offering tariff in LCP is restricted by the imposed price cap $\rho_{k,\max}$ in (21). To clarify the impact of its variation on CO₂ mitigation, a sensitivity analysis is implemented.

Fig. 8 shows that system emissions decrease as the cap value increases, but exhibit saturation after it rises to twice of the initial price. From this curve, it is speculated that an excessively narrow range of prices in RTP could lead to inefficiency for calling upon responses from customers and thus not helpful to CO₂ abatement. This curve could also be used by the government for determining the pricing regulation schemes that properly reflect the impacts of prices on emission reductions.

D. Optimization Performance

The convergence performance and robustness of the optimization is discussed in this section. For this purpose, the simulation is performed for 50 times with random initial data on a Core2 machine with 2.53-GHz CPU and 1-GB RAM, and the OF value (fitness) versus iterations is recorded. The average time for each optimization is 51 min. According to the statistics, the convergence of the best, average and worst simulation (in these 50 times) is shown in Fig. 9.

As is observed, the fitness value decreases sharply in the incipient 50 iterations and stabilizes when 120 improvisations are reached. This demonstrates that the implemented optimization has a satisfactory convergence characteristic with good efficiency. On the other hand, the variation range enclosed by the best and worst individuals (the shadow area) is relatively small in Fig. 9. This indicates that the optimization results are not sensitive to the initial solution, which implies the robustness of the applied method. In this study, although the proposed planning method and algorithm is implemented on a 33-node test system, it should be noted that they are also well applicable to other large

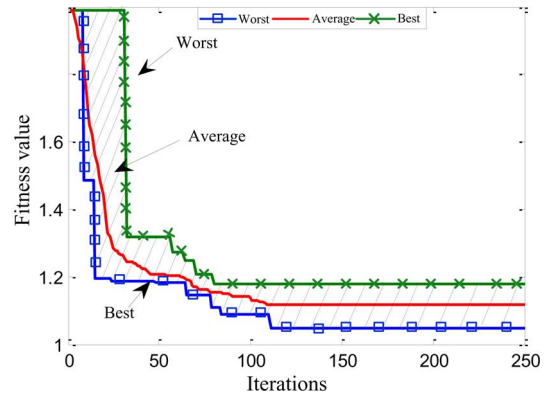


Fig. 9. Convergence performance in the best/average/worst simulation.

systems without difficulty, if needed. In that case, the computation effort will be increased accordingly. Nevertheless, this would not affect the convergence characteristics and robustness as shown in Fig. 9, which manifests that the effectiveness of the algorithm on large scale systems is not rejected.

VIII. CONCLUSION

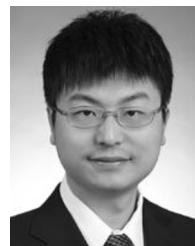
An integrated planning approach for delivering future low-carbon distribution system is proposed in this study. From the DISCO viewpoint, DR and various types of DG are considered as available resources and integrated into network planning for minimizing the total economic and emission cost of the system. As DR enables the electricity consumption to more closely follow the intrinsic production patterns of RDG, the joint planning has demonstrated a superadditive effect in terms of environmental benefits than integrating RDG independently. It is also observed that neglecting the internal DR fading effects and uncertainties in DR may lead to the insufficient network investment. In practice, the actual contribution that LCP could achieve also largely depends on many other factors, such as the imposed caps on tariff fluctuation, and the correlation level between wind speed, solar radiation, load in the target system. In general, more relaxed RTP regulation and higher correlation allow for a more positive effect in CO₂ reduction. This highlights the importance of complete knowledge about system conditions for a successful LCP implementation by DISCOs.

The methodology proposed in this paper is implicitly assumed to be applicable under a central planning context. However, in liberalized electricity markets, this may no longer stand as the private sector could replace DISCOs and be actively engaged in DG investment. In this case, we can build a separate DG planning module from the perspective of private entities and add it into the first-stage to formulate a bi-level game-theory optimization for decision-makings of DSO and DG operators. The objective function and corresponding constraints may need to be adjusted as well in that the network operation efficiency (rather than energy supply costs) would become a prime concern for a DSO on such an occasion. Feed-in tariffs should be put in as a decision variable of the model so as to incentive the DG integration conforming to the benefits of the distribution system. Moreover, as DG and network are managed independently, ancillary services provided by DG would not be free, but compensated through a specialized

market mechanism, the effects of which also requires further exploration. Hence, LCP in the decentralized environment should be a meaningful subject for the future work.

REFERENCES

- [1] IEA CO₂ Emissions From Fuel Combustion, Dec. 2012 [Online]. Available: <http://www.iea.org/w/bookshop/add.aspx?id=618>
- [2] M. Grubb, T. Jamasb, and M. G. Pollitt, *Delivering a Low-carbon Electricity System*. Cambridge, U.K.: Cambridge Univ. Press, 2008.
- [3] P. E. Labisa, R. G. Visandeb, R. C. Pallugnac, and N. D. Caliaoc, "The contribution of renewable distributed generation in mitigating carbon dioxide emissions," *Renew. Sustain. Energy Rev.*, vol. 15, no. 9, pp. 4891–4896, Dec. 2011.
- [4] Y. M. Atwa, E. F. El-Saadany, M. M. A. Salama, and R. Seethapathy, "Optimal renewable resources mix for distribution system energy loss minimization," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 360–370, Feb. 2010.
- [5] D. Q. Hung, N. Mithulananthan, and R. C. Bansal, "Analytical strategies for renewable distributed generation integration considering energy loss minimization," *Appl. Energy*, vol. 105, no. 1, pp. 75–85, May 2013.
- [6] L. F. Ochoa and G. P. Harrison, "Minimizing energy losses: Optimal accommodation and smart operation of renewable distributed generation," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 198–205, Feb. 2011.
- [7] L. F. Ochoa, A. Padilha-Feltrin, and G. P. Harrison, "Time-series-based maximization of distributed wind power generation integration," *IEEE Trans. Energy Convers.*, vol. 23, no. 3, pp. 968–974, Sep. 2008.
- [8] A. Zangeneh, S. Jadid, and A. Rahimi-Kian, "Promotion strategy of clean technologies in distributed generation expansion planning," *Renew. Energy*, vol. 34, no. 12, pp. 2765–2773, Dec. 2009.
- [9] A. Hadian, M.-R. Haghighifard, J. Zohrevand, and E. Akhavan-Rezai, "Probabilistic approach for renewable DG placement in distribution systems with uncertain and time varying loads," in *Proc. 2009 IEEE Power Engineering Society General Meeting*, pp. 1–8.
- [10] S. Wong, K. Bhattacharya, and J. D. Fuller, "Electric power distribution system design and planning in a deregulated environment," *IET Gener. Transm. Distrib.*, vol. 3, no. 12, pp. 1061–1078, Dec. 2009.
- [11] A. Soroudi, R. Caire, N. Hadsaid, and M. Ehsan, "Probabilistic dynamic multi-objective model for renewable and non-renewable distributed generation planning," *IET Gener. Transm. Distrib.*, vol. 5, no. 11, pp. 1173–1182, Nov. 2011.
- [12] M. F. Shaaban, Y. M. Atwa, and E. F. El-Saadany, "DG allocation for benefit maximization in distribution networks," *IEEE Trans. Power Syst.*, vol. 28, no. 2, pp. 639–649, May 2013.
- [13] T. Jin, Y. Tian, C. W. Zhang, and D. W. Coit, "Multicriteria planning for distributed wind generation under strategic maintenance," *IEEE Trans. Power Del.*, vol. 28, no. 1, pp. 357–367, Jan. 2013.
- [14] A. Soroudi and M. Afrasiab, "Binary PSO-based dynamic multi-objective model for distributed generation planning under uncertainty," *IET Renew. Power Gener.*, vol. 6, no. 2, pp. 67–78, Mar. 2012.
- [15] A. J. Roscoe and G. Ault, "Supporting high penetrations of renewable generation via implementation of real-time electricity pricing and demand response," *IET Renew. Power Gener.*, vol. 4, no. 4, pp. 369–382, Jul. 2010.
- [16] R. Sioshansi and W. Short, "Evaluating the impacts of real-time pricing on the usage of wind generation," *IEEE Trans. Power Syst.*, vol. 24, no. 2, pp. 516–524, May 2009.
- [17] S. H. Madaeni and R. Sioshansi, "Using demand response to improve the emission benefits of wind," *IEEE Trans. Power Syst.*, vol. 28, no. 2, pp. 1385–1394, May 2013.
- [18] J. Medina, N. Muller, and I. Roytelman, "Demand response and distribution grid operations: Opportunities and challenges," *IEEE Trans. Smart Grid*, vol. 1, no. 2, pp. 193–198, Sep. 2010.
- [19] P. Mancarella, C. K. Gan, and G. Strbac, "Optimal design of low-voltage distribution networks for CO₂ emission minimisation. Part II: Discrete optimisation of radial networks and comparison with alternative design strategies," *IET Gener. Transm. Distrib.*, vol. 5, no. 1, pp. 47–56, Jan. 2011.
- [20] Z. Liu, F. Wen, and G. Ledwich, "Optimal siting and sizing of distributed generators in distribution systems considering uncertainties," *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2541–2551, Oct. 2011.
- [21] M. G. Lijesen, "The real-time price elasticity of electricity," *Energy Econ.*, vol. 29, no. 2, pp. 249–258, Mar. 2007.
- [22] M. Doostizadeh and H. Ghasemi, "A day-ahead electricity pricing model based on smart metering and demand-side management," *Energy*, vol. 46, no. 1, pp. 221–230, Oct. 2012.
- [23] E. Karangelos and F. Bouffard, "Towards full integration of demand-side resources in joint forward energy/reserve electricity markets," *IEEE Trans. Power Syst.*, vol. 27, no. 1, pp. 280–289, Feb. 2012.
- [24] A. K. David and Y. Z. Li, "Consumer rationality assumptions in the real-time pricing of electricity," *IEE Proc.*, vol. 139, no. 4, pp. 315–322, Jan. 1992.
- [25] M. Zeng, *Theories and Applications of Demand-side Management*. Beijing, China: China Electric Power Press, 2001.
- [26] O. Corradi, H. Ochsenfeld, H. Madsen, and P. Pinson, "Controlling electricity consumption by forecasting its response to varying prices," *IEEE Trans. Power Syst.*, vol. 28, no. 1, pp. 421–429, Feb. 2013.
- [27] Q. Wang, J. Wang, and Y. Guan, "Stochastic unit commitment with uncertain demand response," *IEEE Trans. Power Syst.*, vol. 28, no. 1, pp. 562–563, Feb. 2013.
- [28] W. Yan, F. Liu, C. Y. Chung, and K. P. Wong, "A hybrid genetic algorithm-interior point method for optimal reactive power flow," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1163–1169, Aug. 2006.
- [29] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Trans. Power Del.*, vol. 4, no. 2, pp. 1401–1407, Apr. 1989.
- [30] K. Dietrich, J. M. Latorre, L. Olmos, and A. Ramos, "Demand response in an isolated system with high wind integration," *IEEE Trans. Power Syst.*, vol. 27, no. 1, pp. 20–29, Feb. 2012.
- [31] Technical Guide for Urban Distribution Network, China State Grid Standard Q/GDW 370-2009, Nov. 2009.
- [32] A. M. Smith, Regional Costs of Demand Response Programs in the NEEM Model, Oakland Ridge National Laboratory, Aug. 2011 [Online]. Available: <http://www.spp.gatech.edu/faculty/WOPRpapers/Smith.WOPR-2011.pdf>
- [33] M. Lenzen and J. Munksgaard, "Energy and CO₂ life-cycle analyses of wind turbines—Review and applications," *Renew. Energy*, vol. 26, no. 3, pp. 339–362, Jul. 2002.
- [34] V. M. Fthenakis, H. C. Kim, and E. Alsema, "Emissions from photovoltaic life cycles," *Environ. Sci. Technol.*, vol. 42, no. 6, pp. 2168–2174, Feb. 2008.
- [35] P. J. Meier and G. L. Kulcinski, Life-Cycle Energy Cost and Greenhouse Gas Emissions for Gas Turbine Power, Univ. Wisconsin-Madison, Dec. 2000 [Online]. Available: <http://www.ecw.org/prod/202-1.pdf>
- [36] Z. Qin, W. Li, and X. Xiong, "Incorporating multiple correlations among wind speeds, photovoltaic powers and bus loads in composite system reliability evaluation," *Appl. Energy*, to be published.



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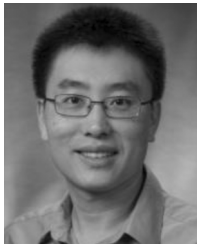
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